38 In an experiment on α-particle scattering, α-particles are directed at a thin gold foil. Most of the α-particles pass straight through the foil or are deflected by a small angle. A small number of α-particles are deflected by a large angle.

Which statement **cannot** be deduced from this experiment?

A Atoms are mostly empty space.

B Most of the mass of an atom is concentrated in the nucleus.

C The nucleus of an atom contains protons.

D The nucleus of an atom is small compared to the size of an atom.